L&T Technology Services

ENGINEERING **THE CHANGE**

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10th Edition TECHgium®

Presentation Round

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GainShield

AI-Driven Safe Bayesian PI Gain Tuning with Stability Certification and Runtime Fallback for Bidirectional EV Charger DC-Link Control



Archana RG



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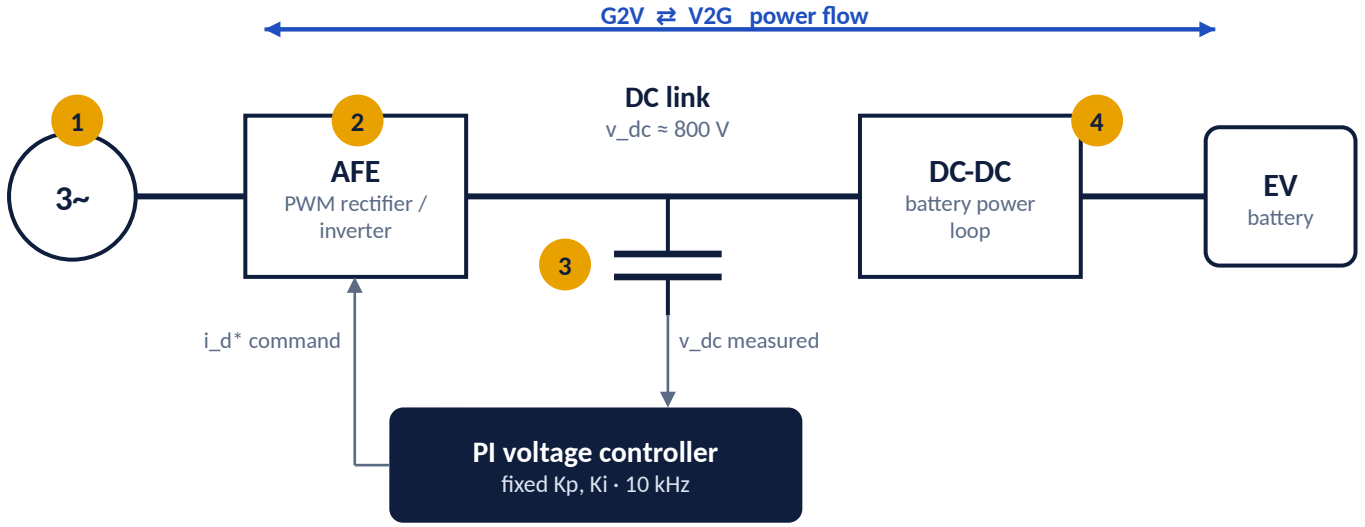
SRM Institute of Science and Technology, Kattankulathur
Chennai, Tamil Nadu

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Challenge Statement

The DC-link loop is tuned once by hand, but the plant it controls changes in every charging session

WHERE THE PLANT MOVES · AVERAGED 30 KW / 800 V MODULE



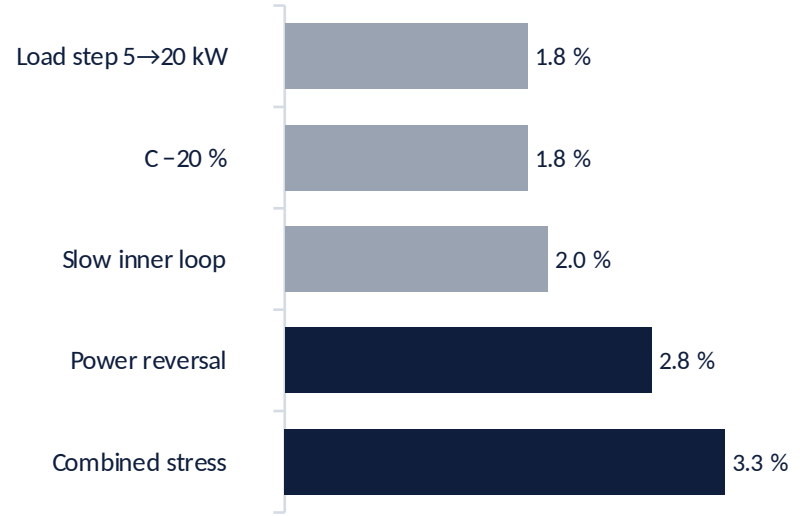
1 **Grid** sags to 0.80 pu; rises to 1.10 pu

2 **Inner loop** lag 0.3–1.0 ms, delay 1–2 samples

3 **Capacitor** C ±20 % with tolerance and ageing

4 **Load** constant-power DC-DC; ±15 kW reversal

EVIDENCE · ONE FIXED PI, FIVE CONDITIONS



Peak DC-link deviation (% of 800 V) of the manually tuned PI, mean of 5 seeds. The same gains give almost twice the deviation under reversal and combined stress.

THE CHALLENGE

Tune online, without ever experimenting blindly on 800 V

Adapt
set Kp, Ki for the current grid, load, capacitance and power direction

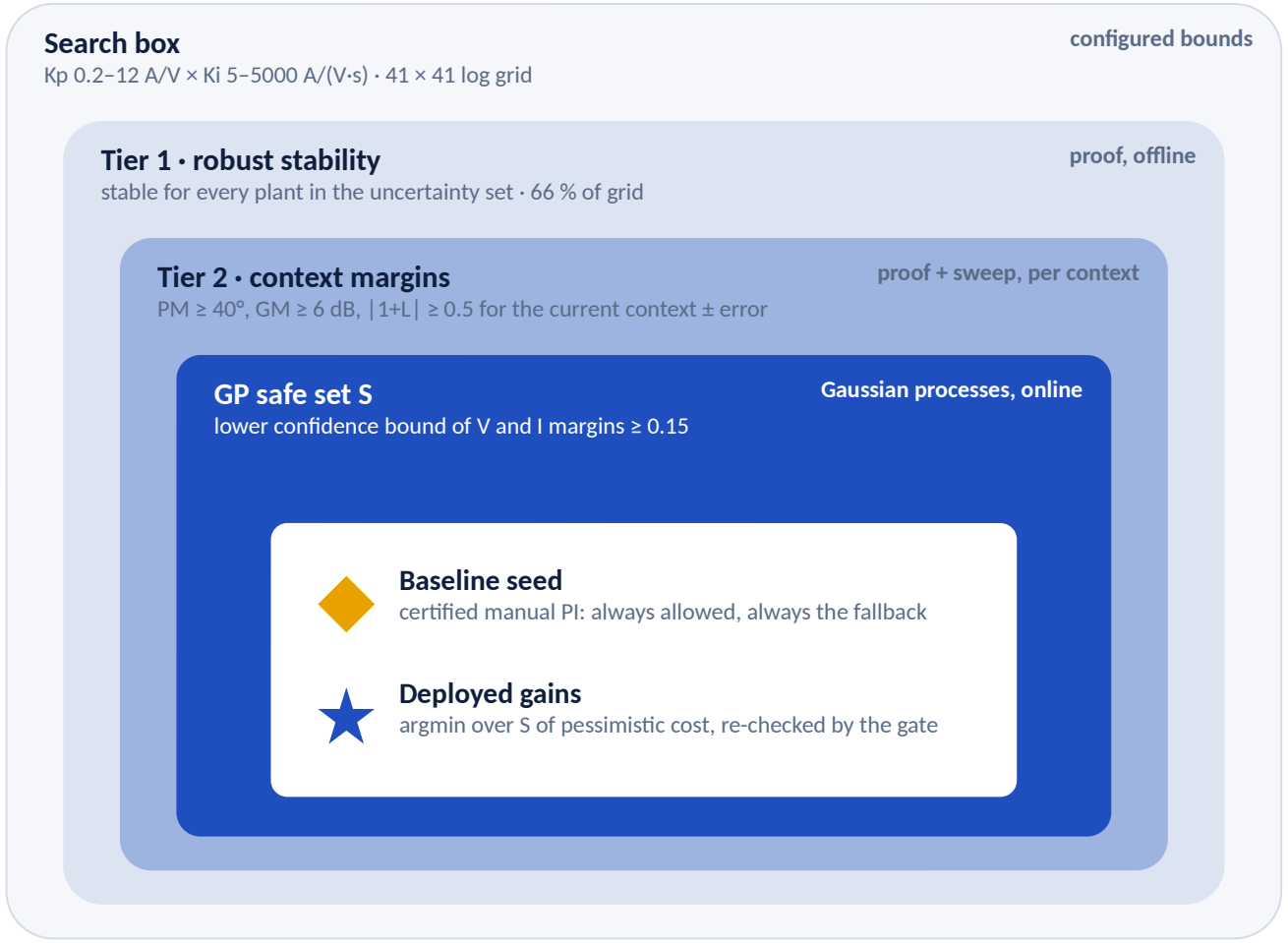
Prove
apply only gains shown stable beforehand; unrestricted BO or RL could try an unstable pair

Fall back · audit
revert instantly to a known-good controller and log every decision

Concept / Proposed Solution

GainShield nests the AI inside a stability proof, which conventional manual or AI tuning cannot offer

THE SAFETY ENVELOPE IN THE KP–KI PLANE



ADVANTAGES OVER CONVENTIONAL TECHNIQUES

	Manual PI	Plain BO / RL	GainShield
Adapts to grid, load, capacitor	✗ no	✓ yes	✓ yes
Proven stable before use	~ one point	✗ no	✓ whole set
Safe while learning on the live loop	– n/a	✗ no	✓ yes
Falls back automatically	✗ no	✗ no	✓ yes
Every gain change audited	✗ no	✗ no	✓ yes
Peak voltage deviation (simulation)	reference	not tested	up to −48 %

1. **Better regulation** up to 48 % smaller voltage swings (G2V → V2G reversal); 23 % on average where it adapts
2. **Safer than plain AI tuning** none of 482 live learning trials left the ±5 % safe band
3. **Accountable** every gain change is logged with its reason and can be replayed

Pros and Cons of the solution

Pros and cons in plain terms: what improves, what it costs, and how each risk is handled

PROS

Smaller voltage swings
When the load or power direction changes, the DC-link voltage swings up to 48 % less than with the hand-tuned controller (23 % on average where it adapts, 19 % across all 11 tests), and recovers sooner.

Never uses an unproven setting
Every gain setting is checked mathematically for stability before it is applied, for all the capacitor, grid and power-direction conditions we model.

Learns safely while running
In 482 live tuning trials, the voltage never left its ±5 % safe band.

Built-in safety net
If the monitor sees anything abnormal, control switches straight back to the proven manual controller, without a jump in the output.

Every decision is recorded
Each gain change is logged with its reason, so any result can be traced, audited and reproduced.

CONS · and how we handle each

Slightly more electrical noise — Handled now
Higher gains make the current command about 14 % noisier.
How we handle it: noise is part of what the tuner minimises, within certified limits

Only as good as the model — Handled now
The stability check assumes our model of the charger is accurate.
How we handle it: a live monitor watches the real signals; hardware tests come next

AI predictions can be wrong — Handled now
The learning model can be confident in conditions it has not tried yet.
How we handle it: it is never trusted alone; uncertain suggestions are rejected

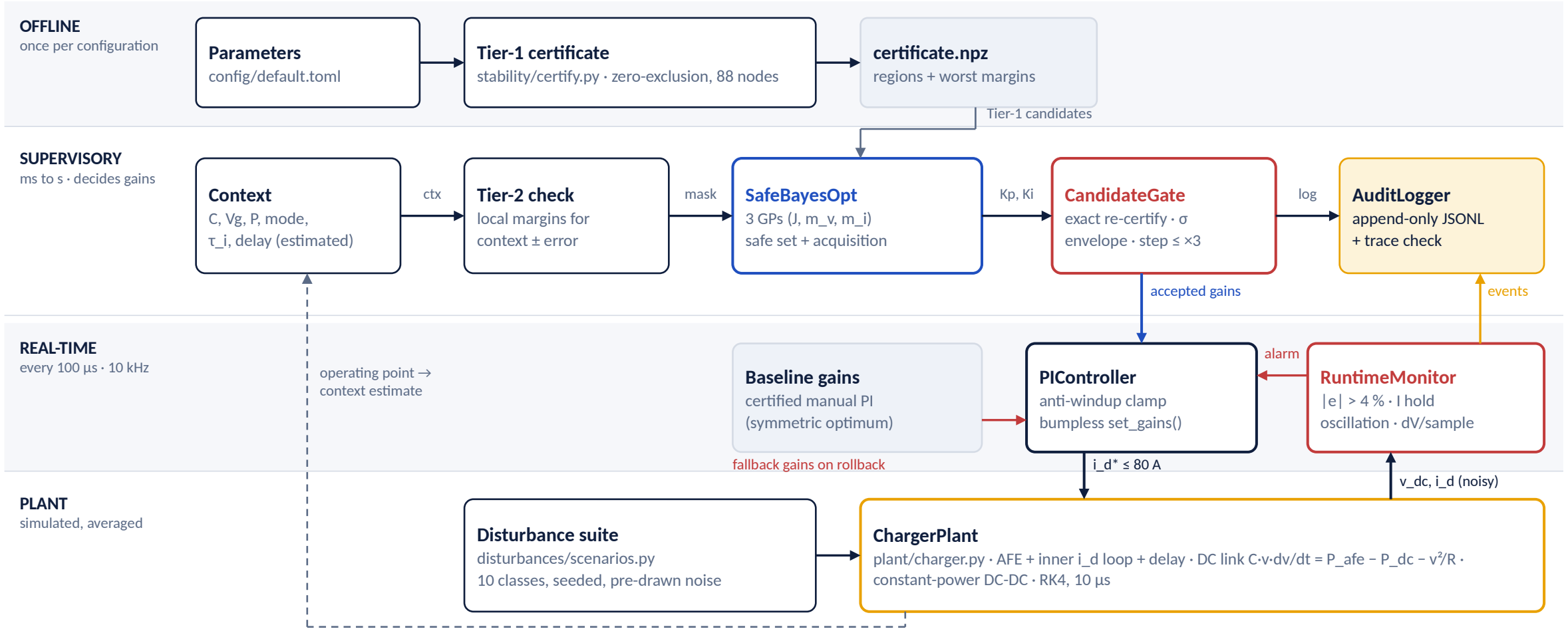
Safety check is slow in Python — Next step
Checking a new operating condition takes about 1.9 s.
How we handle it: it runs outside the fast control loop; a lookup table will remove it

Not yet on real hardware — Next step
Tested in simulation only; sensor checks are tuned on simulated noise.
How we handle it: processor-in-the-loop on a TI board, then lab hardware tests

All figures are from simulation: 11 test conditions × 5 runs, GainShield compared with the hand-tuned controller under identical conditions (results/evaluation_summary.json).

Technical Description

System architecture: four layers separated by time-scale and by trust



Implemented: all four layers run in one Python process (simulation). The lane split is the proposed PC / microcontroller deployment for HIL, not yet built.

Technical Description

Two algorithms: an exact robust-stability test and a safe contextual Bayesian optimiser

A · CERTIFICATE stability/linear_model.py · certify.py

1 Model the sampled charger loop exactly

includes the 1–2 sample delay and inner-loop lag λ = e^(−Ts/τ_i):

$$P(z) = k \cdot \hat{N}(z) / ((z - 1)(z - \lambda)\, z^{(d+1)})$$

2 Collapse all uncertainty into one gain k

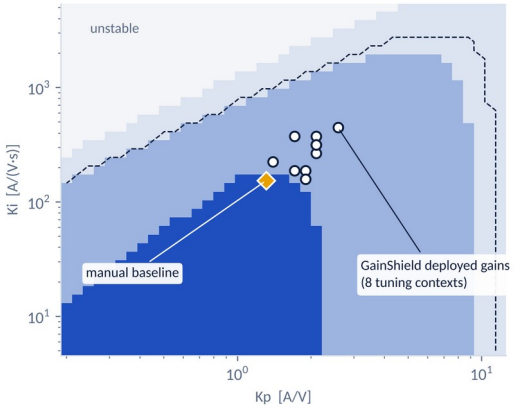
C ±20 %, grid 0.8–1.1 pu and power direction enter only via k ∈ [208, 446]:

$$p(z; k) = D(z) + k \cdot N(z) \qquad \text{affine in } k$$

3 Test every gain pair over the whole k range

if stable at k_min and no k in the range puts a root on |z| = 1, it is stable for all k. Crossings are found as the unit-circle roots of

$$F(z) = D(z)\cdot \text{rev } N(z) - \text{rev } D(z)\cdot N(z)$$



unstable

nominal only

Tier 1 robust · 66 %

globally certified · 22 %

Verified: 100 % agreement with brute-force eigenvalue sweeps; predicts nonlinear instability onset within ±3 % (5/5 cases).

B · SAFE CONTEXTUAL BO, ONE ITERATION optimizer/safe_bo.py

1	z ← context estimate (C, Vg, P, τ_i, delay)	
2	X ← Tier-1 grid ∩ Tier-2(z) ∪ {baseline}	candidates
3	μ, σ ← GP_J, GP_v, GP_i at (X, z)	Matern-5/2
4	**S ← { x : μ_m − β σ_m ≥ h for m ∈ {v, i} }**	safe set
5	M ← { x ∈ S : LCB_J(x) ≤ min_S UCB_J }	maximisers
6	G ← { x ∈ S : UCB_m(x) − L‖x − x′‖ ≥ h, x′ ∉ S }	expanders
7	x* ← argmax over M ∪ G of confidence width	
8	**if Gate(x*) rejects: exclude x*, re-propose**	independent
9	probe episode under RuntimeMonitor → J, m_v, m_i	on plant
10	refit the three GPs; write audit record	

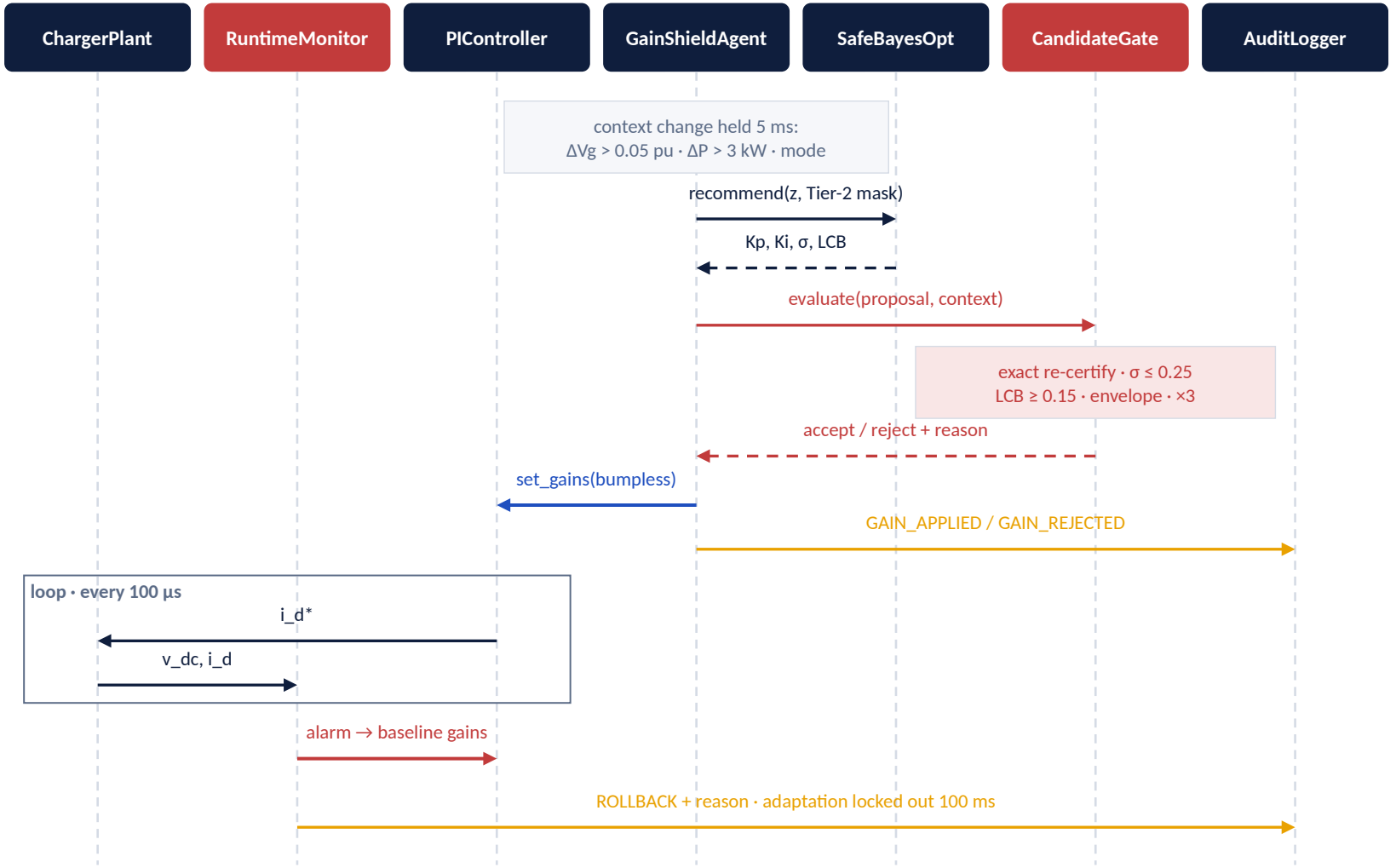
β = 2 · **h = 0.15** · **σ_max = 0.25** · **step ≤ ×3** · **41 × 41 grid** · **zero-prior GPs**

Probe cost J = 0.35 IAE + 0.25 peak dev. + 0.10 overshoot + 0.10 settling + 0.20 ripple (each ÷ baseline).

Follows SafeOpt [1] and its contextual form [2]; their theoretical guarantee is not claimed.

Technical Description

Runtime sequence: every gain change is proposed, re-checked, applied bumplessly and logged



SAFETY AND INTEGRITY, IN DEPTH

1 Tier-1 proof
a context change can never destabilise the active loop

2 Exact re-check
out-of-set proposal rejected (test G1: yes)

3 GP σ and LCB
extrapolation rejected (G2, G3)

4 Envelope + step
plant outside model, large jumps rejected (G4, G5)

5 Monitor + fallback
unstable gains reaching the plant rolled back (R1)

6 Sensor plausibility
stuck sensor caught; a power-balance estimate takes over (R5)

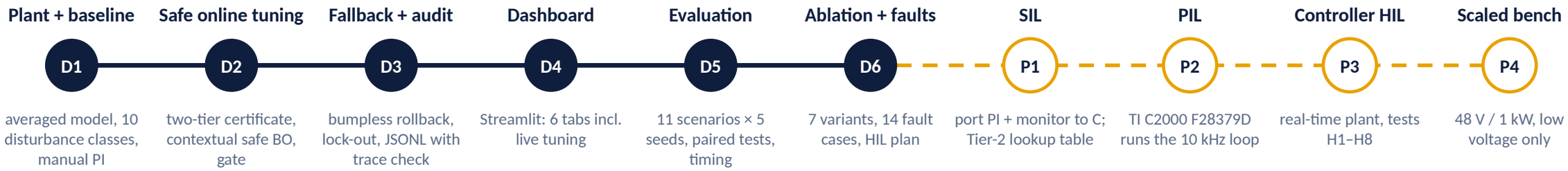
7 Audit trace check
every gain change explained by a record

Last line independent hardware over/under-voltage trip behind every software layer (HIL roadmap).

Implementation Plan

All six deliverables are implemented and tested in simulation; hardware-in-the-loop is the next phase

IMPLEMENTED AND TESTED IN SIMULATION | **PROPOSED · NOT YET EXECUTED**



REQUIREMENT TRACEABILITY summary of docs/REQUIREMENTS_TRACEABILITY.md

Deliverable	Module	Test	Evidence	Status
D1 Plant, constraints, disturbances, baseline	plant/, disturbances/, controllers/	test_plant, test_scenarios	energy balance, eval figures	Complete and tested (sim)
D2 Safe online tuning + monitors	stability/, optimizer/, safety/	test_stability, _optimizer, _supervisor	0 of 482 trials unsafe; stuck sensor caught	Complete and tested (sim)
D3 Fallback, rollback, audit	core.py, audit/logger.py	test_integration, test_audit	0 trace problems	Complete and tested (sim)
D4 Dashboard	dashboard/app.py	headless AppTest	6 tabs, no exceptions	Complete and tested (sim)
D5 Evaluation	experiments/run_evaluation.py	55 paired runs	evaluation_summary.json	Complete and tested (sim)
D6 Ablation, failures, HIL roadmap	run_ablation.py, run_failure_injection.py	7 variants, 14 fault cases	HIL_ROADMAP.md: 8 tests, costed	Complete and tested (sim)

Implementation Plan

Implementation: a tested Python package, a reproducible pipeline and a live dashboard

ONE COMMAND REPRODUCES EVERY NUMBER

1. **run_certificate.py** ≈ 20 s
   → certificate.npz
2. **run_tuning.py** 13 contexts
   → model.pkl · tuning_audit.jsonl
3. **run_evaluation.py** 55 pairs
   → evaluation_summary.json
4. **run_failure_injection.py** 14 cases
   → failure_injection.csv
5. **run_ablation.py** 7 variants
   → ablation.csv
6. **make_reports.py**
   → docs/*.md · RESULTS_SNAPSHOT.json
7. **build_deck.py**
   → this presentation

python experiments/run_all.py

≈ 30 min on a 2-core CPU; fixed seeds; no GPU, MATLAB or network.

TECHNOLOGY STACK (AS IMPLEMENTED)

FRONTEND	**Streamlit + Plotly dashboard** dashboard/app.py · monitoring, AI, safety, experiments, audit, live tuning
EXPERIMENTS	**pandas · Matplotlib · SciPy stats** experiments/ · reports generated, not typed
CORE PACKAGE	**NumPy · scikit-learn** gainshield/ · plant RK4, certificate, 3 GPs, supervisor, audit
QUALITY	**pytest · Streamlit AppTest** 40 unit + integration tests; dashboard tested headless
RUNTIME	**Python ≥ 3.11 · CPU only** pinned versions in requirements-lock.txt

Why not PyTorch / BoTorch: the safe-set logic stays auditable line by line and runs on a CPU in seconds, without a ~1 GB dependency.

PROJECT RESOURCES

Scan: GitHub repository
github.com/Thiyagadurden/GainShield

README with results and quick start

Model, architecture, assumptions

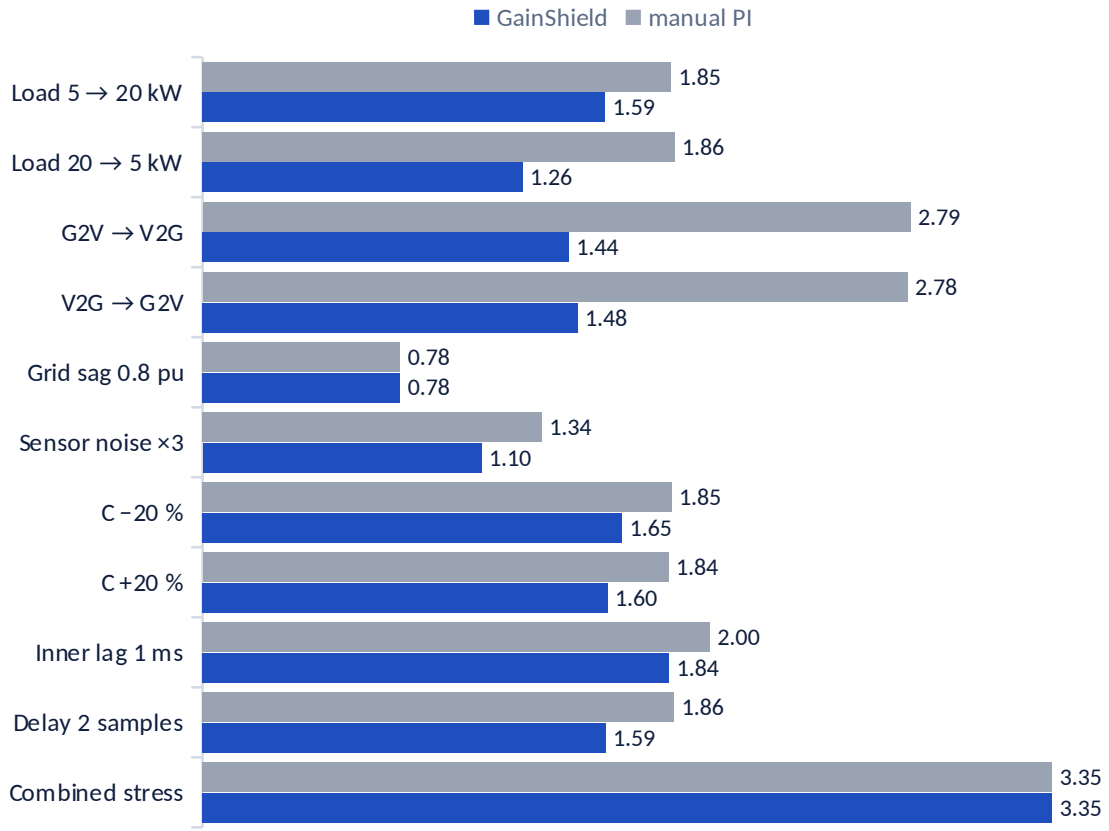
Traceability, HIL roadmap, costs, limitations

Generated results, figures, audit logs

Validation / Testing / Analysis

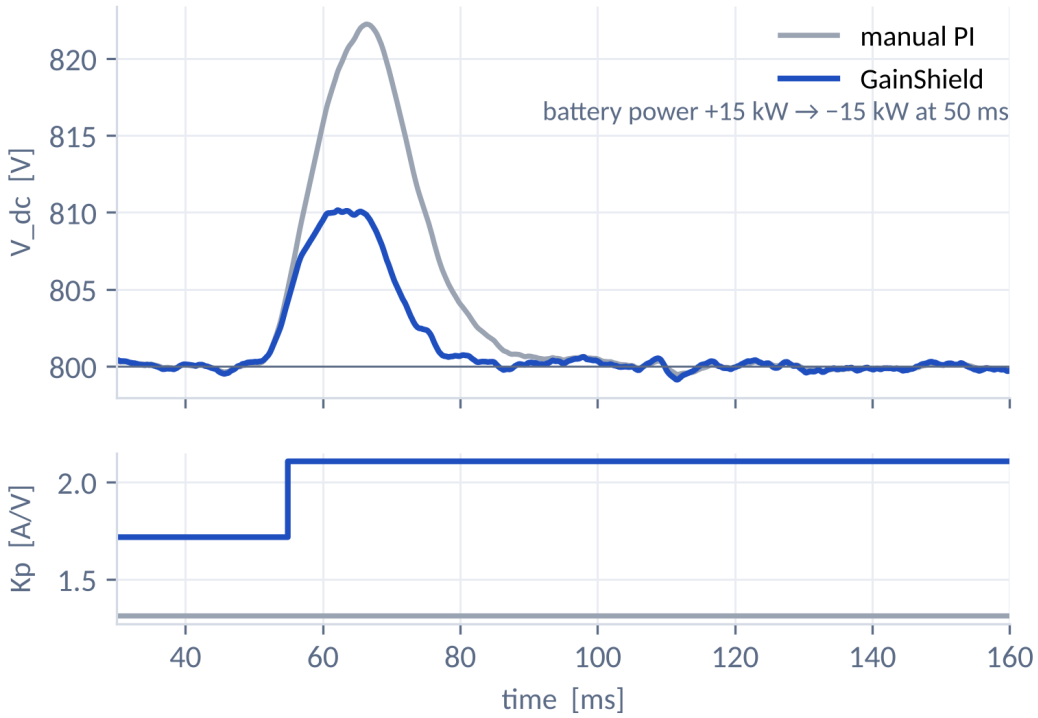
Up to 48 % smaller voltage swings: better in 9 of 11 scenarios and never worse

PEAK DC-LINK DEVIATION, % OF 800 V · MEAN OF 5 SEEDS



Protocol 11 scenarios × 5 seeds; identical plant, events, pre-drawn noise and context error; Ts = 100 µs; metrics on true plant states. Sag, slow lag and combined: GainShield keeps the baseline.

POWER REVERSAL +15 → −15 KW · SEED 0



−48 %
peak deviation, G2V → V2G reversal

−23 %
average where it adapts (9/11)

−19 %
average over all 11 scenarios

0
voltage, current or trip violations

Source: results/evaluation_summary.json · Wilcoxon p = 7.6e-09 (peak deviation, 55 pairs) · reversal peak 2.79 % → 1.44 %

Validation / Testing / Analysis

Ablation and fault injection: removing a safety layer turns a contained fault into a sustained oscillation

REMOVE ONE LAYER AT A TIME · SAME SEEDS, SAME FAULT

Variant	Unsafe tuning episodes	Deployed peak vs manual PI	Injected unstable gains: peak · time outside ±5 %
A Complete GainShield	0	0.80 ×	3.9 % · 0 ms
B – certified region	0	0.86 ×	3.9 % · 0 ms
C – GP uncertainty	0	0.69 ×	4.7 % · 0 ms
D – runtime monitor	0	0.80 ×	**7.5 % · 48 ms**
E – context	0	0.89 ×	4.7 % · 0 ms
F – fallback	0	0.80 ×	**7.5 % · 48 ms**

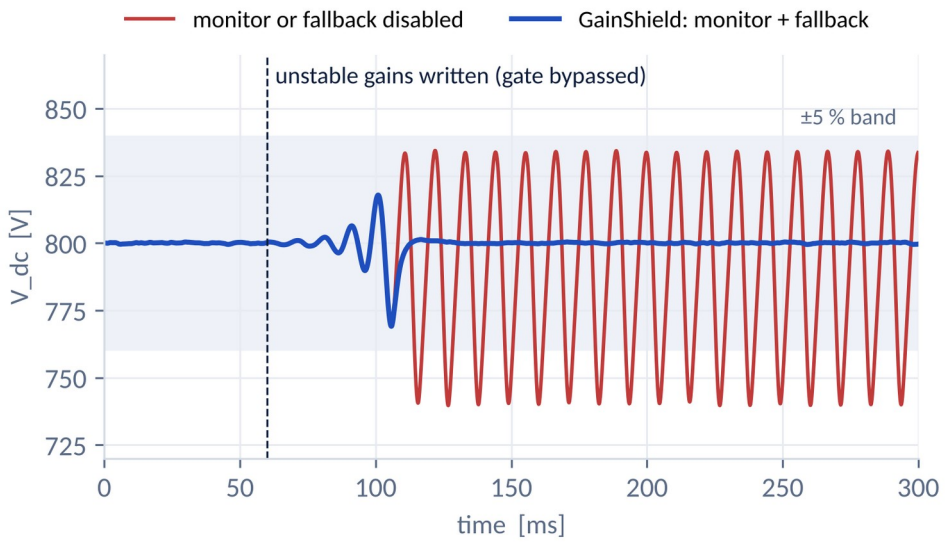
D, F without the monitor or the fallback the fault persists as a limit cycle; with them it is rolled back.

C trusting GP means gives faster but unverified gains (only 35 measured episodes).

B no unsafe episode in this budget: the certificate's value is a guarantee, not an average.

E without context the gain shrinks (0.89 × vs 0.80 ×).

FAULT: UNSTABLE GAINS WRITTEN BEHIND THE GATE



5 / 5

unsafe proposals rejected by the gate

46 ms

injection to rollback (4 % abort band)

Stuck sensor (R5): closed

caught in 1.9 ms, voltage held within 1.3 % on the observer estimate. Without the check: 96 % collapse, trip.

Source: results/ablation.csv, results/failure_injection.csv. Isolated simulation; unsafe variants are never for hardware.

Cost Estimate

Investment roadmap: about ₹20,000 takes GainShield onto real controller hardware

1 Processor-in-the-loop

₹14–21k

estimate

2 × TI LAUNCHXL-F28379D	**₹8.3–11.4k**
signal conditioning, isolation	**₹6–10k**

Unlocks
10 kHz PI + monitor on a real MCU; measured delay and timing

2 Controller HIL · institute lab

₹0–50k

usage charges, if any

real-time plant simulator access	**₹0–50k**

Unlocks
fault tests H1–H8, incl. sensor faults and comms loss

3 Scaled power bench

₹1.5–3 L

engineering estimate

48 V / 1 kW converter bench	**₹1.5–3 L**

Unlocks
real switching, sensors and losses at safe voltage

4 Own HIL platform · optional

₹9.3–41 L

only if lab access is unavailable

Typhoon HIL101, MSRP $9,900	**≈ ₹9.3 L**
HIL404 class, quote $44k	**≈ ₹41 L**

Unlocks
dedicated, repeatable HIL regression

WHERE THE NEXT IMPROVEMENT COMES FROM

−48 %

best case today
peak deviation, G2V → V2G reversal; −18.7 % mean over 55 paired runs

≈ −13 %

further headroom observed
unverified GP-extrapolated gains reached 0.69× vs 0.80× deployed (ablation); more measured episodes per context can recover part of it safely

1.9 s

decision latency to remove
precomputed Tier-2 lookup table instead of the Python check

+14 %

noise penalty to win back
current-command ripple vs manual PI; ripple-aware cost weighting or a notch filter

Prices researched 12 Nov 2026, 1 USD ≈ ₹94 (Arrow, Typhoon HIL, Indian distributor listings); budgetary, no vendor quotation, no partnership; foreign prices exclude duties/GST. L = lakh. Headroom figures: results/ablation.csv (simulation).

Assumptions

No numbers were supplied, so every value is a stated, configurable engineering assumption

PLANT

A-1/2	30 kW module, 800 V DC link
A-3	C = 2 mF, ±20 %
A-4	415 V grid, 0.80–1.10 pu
A-6/7	AFE η = 98 %; inner loop τ_i 0.3–1.0 ms
A-8	DC-DC τ = 2 ms, 2 kW/ms, ±30 kW

LIMITS

A-9/10	10 kHz switching = sampling; delay 1.0–2.0 samples
A-11/12	i_d ≤ 80 A; ±5 % band, ±10 % trip, 4 % abort
A-14	sensor noise 1 V, 0.5 A (1σ)

STABILITY

A-13	worst case PM ≥ 40°, GM ≥ 6 dB, |1+L| ≥ 0.5
model	linearised, unsaturated; lag × delay gridded (88 nodes)

OPTIMISER

A-15	context known within C ±5 %, τ_i ±20 %, delay ±0.2
BO	β = 2, h = 0.15, σ ≤ 0.25, step ≤ ×3; RKHS bound unverified
probe	+10 V reference step and ±6 kW power step acceptable

BOUNDARY

scope	averaged model: no switching ripple, AC filter, PLL, battery, thermal
tests	simulation only: no SIL, HIL or hardware

All values live in config/default.toml with rationale in docs/ASSUMPTIONS.md; changing them and running run_all.py regenerates every result.

References

References: each supports a specific design decision

SAFE BAYESIAN OPTIMISATION

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ENGINEERING **THE CHANGE**

Purposeful.
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Thank You

GainShield

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Code, results and documentation
github.com/Thiyagadurden/GainShield

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